Registration No.:_____

IMPORTANT NOTE

1) This sample paper serves solely as a visual representation of the question paper.
2) Each question in the question paper is mapped with corresponding Course Outcome(s) (CO) and Revised Bloom's Taxonomy (RBT) Level. However the question-wise CO-RBT mapping given below is indicative only and will vary in the final question paper.
3) Students should refer to the latest course syllabus and question paper pattern for exam preparation.
4) The final version of the End Term Examination question paper may differ in terms of layout, design, and content. This sample paper should be viewed for reference purpose only and not as final/official question paper.

23242CSE58048
Paper Code: A

Course Code:CSE357
Course Title:COMBINATORIAL STUDIES

Time Allowed: 03:00hrs. **Max Marks: 60**

Read the following instructions carefully before attempting the question paper.
1. Match the Paper Code shaded on the OMR Sheet with the Paper code mentioned on the question paper and ensure that both are the same.
2. This question paper contains 60 questions of 1 mark each. 0.25 marks will be deducted for each wrong answer.
3. Attempt all the questions in serial order.
4. Do not write or mark anything on the question paper and/or on rough sheet(s) which could be helpful to any student in copying, except your registration number on the designated space.
5. Submit the question paper and the rough sheet(s) along with the OMR sheet to the invigilator before leaving the examination hall.

Q(1) What characteristic defines a graph as a weighted graph?
(a) It has nodes and edges.
(b) It has edges with weights assigned to them.
(c) It has a cycle.
(d) It has a root node.

,L2

Q(2) Which of the following data structure can't store the non-homogeneous data elements?
(a) Arrays
(b) Records
(c) Pointers
(d) Stacks

CO1,L1

Q(3) What is the primary role of a compiler in the programming process?
(a) To execute code line by line and provide immediate feedback.
(b) To translate high-level code into machine code all at once.
(c) To allocate memory for variables during runtime.
(d) to provide an interactive environment for testing code.

,L2

Q(4) In the context of data integrity, evaluate the role of a primary key in a relational database table.
(a) It allows duplicate values in a column to be indexed for faster retrieval.
(b) It establishes a unique identifier for each row, ensuring no two rows have the same value in this column.
(c) It permits NULL values in the column for flexibility in data entry.
(d) It links tables together to enforce referential integrity.

,L5

Q(5) Dynamic programming strategies involve:
(a) Solving subproblems multiple times
(b) Memorizing solutions to subproblems
(c) Ignoring subproblems with low values
(d) Random selection of subproblems

CO1,L6

Q(6) What is the full form of OSI?
(a) optical service implementation
(b) open service Internet
(c) open system interconnection
(d) operating system interface

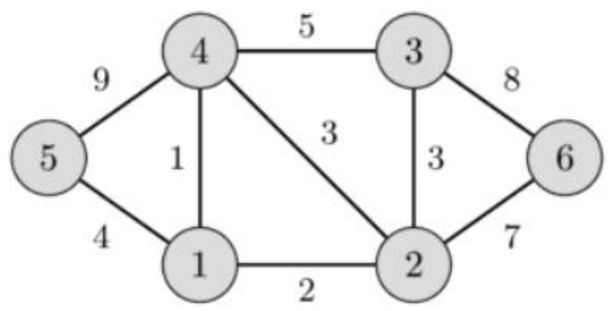
CO4,L6

Q(7) Identify scheduling algorithm ensures fairness by assigning each process an equal time slice?
(a) First-Come, First-Served (FCFS)
(b) Shortest Job First (SJF)
(c) Round Robin (RR)
(d) Priority Scheduling

CO1,L2

Q(8)

Consider the following undirected graph with edge weights as shown:



The total weight of minimum spanning tree of the graph is _____

(a) 20
(b) 18
(c) 17
(d) 10

CO5,L5

Q(9) Indicate the purpose of a Memory Management Unit (MMU)?
(a) To allocate memory to processes
(b) To manage virtual memory
(c) To translate logical addresses to physical addresses
(d) To swap processes in and out of memory

CO1,L2

Q(10) Concurrent processes are processes that _____?
(a) Overlap in time
(b) Do not overlap in time
(c) Are executed by a processor at the same time
(d) None of the above

CO1,L2

Q(11) Explain which of the following is not a restriction for a table to be a relation?
(a) The cells of the table must contain a single value
(b) All of the entries in any column must be of the same kind
(c) The columns must be ordered
(d) No two rows in a table may be identical

CO4,L2

Q(12) What is the primary reason why relational databases avoid storing duplicate data?
(a) To reduce disk storage costs
(b) To improve data integrity and consistency
(c) To make queries faster
(d) To allow more concurrent users

CO4,L4

Q(13)

Identify Consider a system with three processes scheduled using the Round Robin (RR) scheduling algorithm. The time quantum is set to 4 milliseconds. The arrival time and burst time for each process are as follows:

Process	Arrival Time	Burst Time
P 1	0	8
P 2	2	5
P 3	4	6

If the processes follow the FCFS order when multiple processes have the same remaining time, what is the turnaround time of process P2?

(a) 15
(b) 16
(c) 14
(d) 18

CO1,L2

Q(14) What is the return value of the fork system call in the parent process?
(a) -1
(b) 0
(c) Process ID (PID) of the child process
(d) Process ID (PID) of the parent process

CO1,L2

Q(15) A connectionless protocol that provides a simple and faster way to transmit data, often used for real-time application is
(a) DNS
(b) TCP
(c) IP
(d) UDP

CO2,L5

Q(16) What is the output of the following code snippet?
#include using namespace std; int main() { int *ptr = NULL; cout << *ptr << endl; return 0; }
(a) Segmentation fault
(b) Runtime error
(c) 0
(d) Garbage value

CO1,L1

Q(17) The value of the following expression
(13/4*3)% 5 + 1) is
(a) 5.75
(b) 1.4875
(c) 3.95
(d) 5

CO3,L1

Q(18) Which of the following is TRUE about a Compiler?
(a) It converts high-level code to machine code line-by-line
(b) It uses an interpreter internally
(c) It directly executes the high-level program
(d) It converts high-level code to machine code all at once

CO1,L3

Q(19) Storage organization refers to:
(a) Arranging files on disk
(b) Allocating memory for variables
(c) Sorting data in memory
(d) Formatting storage devices

CO1,L1

Q(20) Tell Which of the following is NOT a principle of data structure design?
(a) Efficiency
(b) Consistency
(c) Modularity
(d) Flexibility

CO1,L1

Q(21) To perform level-order traversal on a binary tree, which of the following data structure will be required?
(a) Hash table
(b) Queue
(c) Binary search tree
(d) Stack

CO1,L6

Q(22) In graph traversal, what is the main difference between depth-first search (DFS) and breadth-first search (BFS)?
(a) DFS uses a queue, while BFS uses a stack.
(b) DFS explores as far as possible along each branch before backtracking, while BFS explores neighbors before moving on.
(c) DFS guarantees the shortest path between two nodes, while BFS does not.
(d) BFS uses recursion, while DFS uses iteration.

CO6,L4

Q(23) Which of the following is a valid max-heap property?
(a) Parent node is always greater than or equal to its children
(b) Parent node is always less than or equal to its children
(c) Every level of the tree is fully filled
(d) Heap follows an inorder traversal structure

CO5,L2

Q(24) What will the following SQL query do?
SELECT * FROM employees WHERE department = 'Sales';
(a) Select all columns from the employees table where the department is 'Sales'.
(b) Select the department column from the employees table where the department is 'Sales'.
(c) Select the first employee from the employees table where the department is 'Sales'.
(d) Select the count of employees in the 'Sales' department.

CO1,L3

Q(25) What is the role of the CHECK constraint in a CREATE TABLE statement?
(a) Ensures the column is not null
(b) Enforces a condition on the values allowed in a column
(c) Specifies a default value for a column
(d) Sets the auto-increment property for a column

CO3,L3

Q(26) Decide What is the maximum data rate for a Category 6 Ethernet cable?
(a) 100 Mbps
(b) 1 Gbps
(c) 10 Gbps
(d) 100 Gbps

CO1,L5

Registration No.:_____

Q(27) In heap sort, after deleting the last minimum element, the array will contain elements in?
(a) increasing sorting order
(b) tree preorder
(c) tree inorder
(d) decreasing sorting order

CO1,L6

Q(28) In which network topology is the failure of one node least likely to affect the rest of the network?
(a) Bus
(b) Star
(c) Ring
(d) Mesh

CO3,L4

Q(29) In the context of data integrity, evaluate the role of a primary key in a relational database table.
(a) It allows duplicate values in a column to be indexed for faster retrieval.
(b) It establishes a unique identifier for each row, ensuring no two rows have the same value in this column.
(c) It permits NULL values in the column for flexibility in data entry.
(d) It links tables together to enforce referential integrity.

,L5

Q(30) A class C network 192.168.1.0/24 is divided into subnets of size 32 IP addresses. How many subnets are created?
(a) 4
(b) 8
(c) 16
(d) 64

CO3,L4

Q(31) What is the role of the linker in the compilation process?
(a) To interpret source code
(b) To resolve external references and combine object files
(c) To optimize the machine code
(d) To debug the program

CO5,L4

Q(32) Identify a key advantage of cloud-based databases compared to traditional on-premises databases?
(a) Automatic scaling based on demand
(b) Greater control over physical hardware
(c) Higher security risks due to internet exposure
(d) Inability to support high availability

CO4,L2

Q(33) Which of the following statements is true regarding Big-O notation?
(a) It provides an exact measure of the algorithm's running time.
(b) It describes the best-case scenario of an algorithm.
(c) It focuses on the upper bound of the growth rate of an algorithm.
(d) It is only applicable to recursive algorithms.

CO5,L1

Q(34) What does a system call provide in an operating system?
(a) A way for programs to request services from the operating system
(b) User interface
(c) Hardware resources
(d) File management

CO1,L2

Q(35) What is the purpose of a composite key?
(a) To establish relationships between tables
(b) To uniquely identify each record in a table
(c) To prevent duplicate values in a column
(d) To represent a combination of multiple attributes as a single key

CO1,L3

Q(36) In a Priority Queue, the element with the highest priority is always _____
(a) Inserted at the front
(b) Dequeued first
(c) Inserted at the rear
(d) Never dequeued

CO5,L3

Q(37) Consider a situation where a function needs to modify the value of a variable passed to it. Which parameter passing technique would you choose to achieve this?
(a) Pass by value
(b) Pass by reference
(c) Pass by constant reference
(d) Pass by copy

,L3

Registration No.:_____

Q(38) What is the difference between INNER JOIN and LEFT JOIN in SQL?
(a) INNER JOIN returns only matching rows, while LEFT JOIN returns all rows from the left table and matching rows from the right table.
(b) INNER JOIN returns all rows from both tables, while LEFT JOIN returns only matching rows.
(c) INNER JOIN and LEFT JOIN are identical operations in SQL.
(d) INNER JOIN returns all rows from the left table and matching rows from the right table, while LEFT JOIN returns only matching rows.

CO3,L3

Q(39) Demonstrate for an effective operating system, when to check for deadlock?
(a) every time a resource request is made at fixed time intervals
(b) at fixed time intervals
(c) every time a resource request is made
(d) none of the mentioned

CO1,L3

Q(40) Choose What is the time complexity of the Floyd-Warshall algorithm for finding the shortest paths between all pairs of vertices in a weighted graph?
(a) O(V)
(b) O(V^2)
(c) O(E log V)
(d) O(V^3)

CO1,L6

Q(41) Which queue allows insertion and deletion from both ends?
(a) Circular Queue
(b) Priority Queue
(c) Deque
(d) Simple Queue

CO5,L2

Q(42) What is the primary limitation of Dijkstra's algorithm when dealing with graphs containing negative-weight edges?
(a) Dijkstra's algorithm becomes slower.
(b) Dijkstra's algorithm may produce incorrect results.
(c) Dijkstra's algorithm is not affected by negative-weight edges.
(d) Dijkstra's algorithm cannot handle graphs with negative-weight edges.

CO6,L4

Q(43) In Quick Sort, what is the element called around which the partitioning takes place?
(a) Pivot
(b) Key
(c) Seperator
(d) Divider

CO1,L6

Q(44) When designing a data structure that needs to dynamically grow and shrink, which underlying storage mechanism would you typically use?
(a) Array
(b) Linked List
(c) Static Buffer
(d) Fixed-size Array

,L3

Q(45) Pointers in C/C++ are used to:
(a) Store character data
(b) Manage memory addresses
(c) Control program flow
(d) Perform input/output operations

CO4,L1

Q(46) Which of the following is NOT a common application of queues?
(a) Printer job scheduling
(b) CPU scheduling
(c) Undo operation in software
(d) Handling requests in a server

CO2,L4

Q(47) What is the purpose of algorithm analysis?
(a) To determine the best algorithm for a given problem
(b) To measure the efficiency of an algorithm
(c) To identify errors in an algorithm
(d) All of the above

CO4,L1

Q(48) Which type of network covers a limited geographical area such as a building or campus?
(a) LAN (Local Area Network)
(b) WAN (Wide Area Network)
(c) MAN (Metropolitan Area Network)
(d) PAN (Personal Area Network)

CO1,L5

Registration No.:_____

Q(49) Which of the following scenarios will lead to a violation of 4NF (Fourth Normal Form) in a relational database?
(a) A relation where a non-prime attribute is transitively dependent on a candidate key.
(b) A relation where an attribute is dependent on more than one candidate key.
(c) A relation where an attribute is multivalued, and it depends on another non-prime attribute.
(d) A relation where a multivalued dependency exists.

CO4,L6

Q(50) The primary purpose of an operating system is to:
(a) Manage hardware resources
(b) Develop software applications
(c) Enhance user interfaces
(d) Perform data analysis

CO1,L2

Q(51) Which memory management technique divides physical memory into fixed-sized blocks to store processes?
(a) Paging
(b) Fragmentation
(c) Segmentation
(d) Swapping

CO1,L2

Q(52) PoP stand for?
(a) Pre Office Protocol
(b) Post Office Protocol
(c) Protocol of Post
(d) None

CO1,L5

Q(53) Determine that is network layer at source is responsible for creating a packet from data coming from another _____
(a) Station
(b) Link
(c) Node
(d) Protocol

CO3,L3

Q(54) Choose Which of the following is an advantage of adjacency list representation over adjacency matrix representation of a graph?
(a) In adjacency list representation, space is saved for sparse graphs.
(b) DFS and BSF can be done in O(V + E) time for adjacency list representation. These operations take O(V^2) time in adjacency matrix representation. Here is V and E are number of vertices and edges respectively.
(c) Adding a vertex in adjacency list representation is easier than adjacency matrix representation.
(d) All of the above

CO1,L6

Q(55) HASH functions are used to produce the message digests which are then encrypted with a private key to get :
(a) Public key
(b) Digital signature
(c) Cipher text
(d) Data Encryption Standard

CO4,L6

Q(56) Which queue allows insertion at one end and deletion at both ends?
(a) Circular Queue
(b) Deque
(c) Priority Queue
(d) Simple Queue

CO5,L5

Q(57) In classful addressing, a large part of available addresses are _____
(a) Wasted
(b) Blocked
(c) Organized
(d) Communicated

CO1,L5

Q(58) Which of the following search algorithms does not require collection to be sorted first?
(a) Linear Search
(b) Interpolation Search
(c) Binary Search
(d) None of the above

CO2,L3

Q(59) Which of the following is TRUE about the 'register' storage class?
(a) It always stores variables in CPU registers
(b) It reduces memory access time
(c) It can be accessed using pointers
(d) It has a global scope

CO5,L3

Q(60) Choose Which sorting algorithm exhibits the best performance for nearly sorted arrays?
(a) Merge Sort
(b) Bubble Sort
(c) Quick Sort
(d) Insertion Sort

CO1,L6

--End of Question paper--